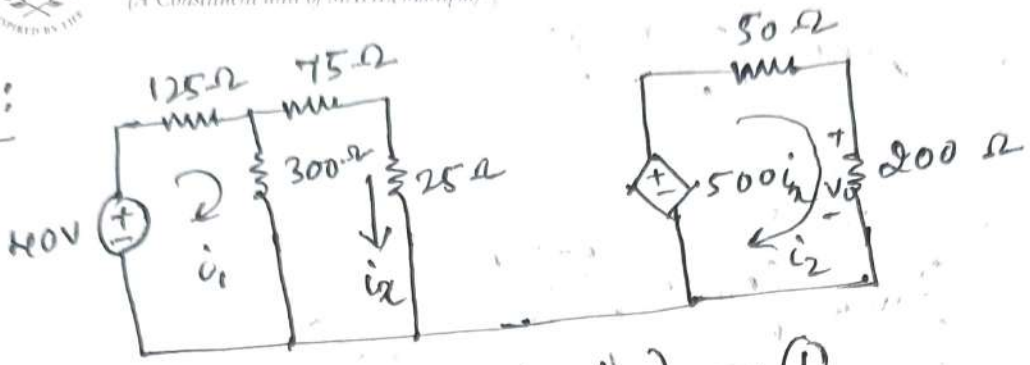
Q1:



$40 - 125i_1 - 300(i_1 - i_x) \quad \text{—— (1)}$

$75i_x + 25i_x + 300(i_x - i_1) = 0 \quad \text{—— (2)}$

$50i_2 + 200i_2 - 500i_x = 0 \quad \text{—— (3)}$

$\therefore \quad i_1 = 0.2A \; ; \; i_2 = 0.3A \; ; \; i_x = 0.15A. \quad \text{—— (3M)}$

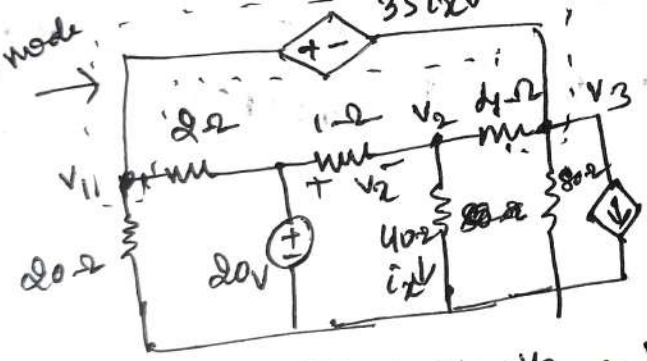
$V_0 = 200 \times i_2 \Rightarrow V_0 = 200 \times 0.3 = 60V$

Power delivered by dependent source $= -(500 i_x) i_2$

$= -500(0.15)(0.3)$

$= 22.5\,W \quad \text{—— (1M).}$

Q2:



$\frac{V_1}{20} + \frac{V_1 - 20}{2} + \frac{V_3 - V_2}{4} + \frac{V_3}{80} + 3.125\,V_x = 0$

$\frac{V_2}{40} + \frac{V_2 - V_3}{4} + \frac{V_2 - 20}{1} = 0 \quad \text{—— (1M)}$

but $V_x = 20 - V_2 \quad (20 = V_x + V_2)$

$V_1 - 35i_x = V_3$

$i_x = \frac{V_2}{40} \quad \text{—— (1M)}$

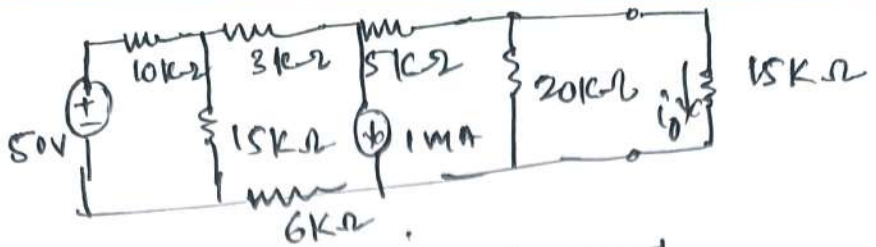
$\therefore \quad V_1 = -20.25\,V \; ; \; V_2 = 10V \; ; \; V_3 = -29\,V.$

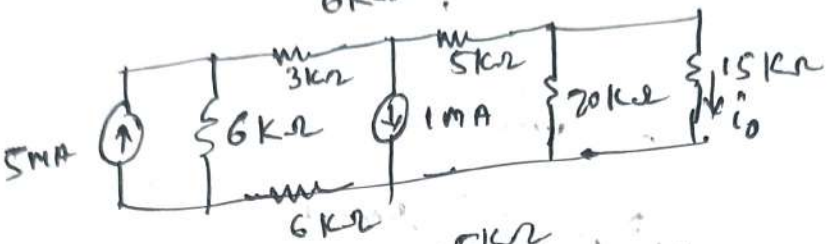
Current across 20V is

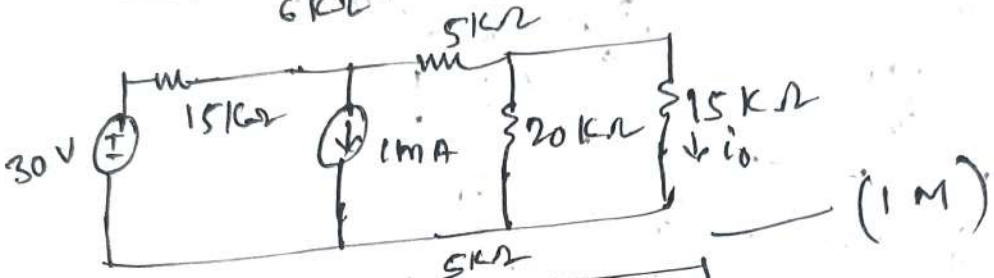
$I = \frac{20 - 10}{1} + \frac{20 - (-20.25)}{2} = 30.125\,A$

$P = 20 \times 30.125 = 602.5\,W \quad \text{(1M)}$

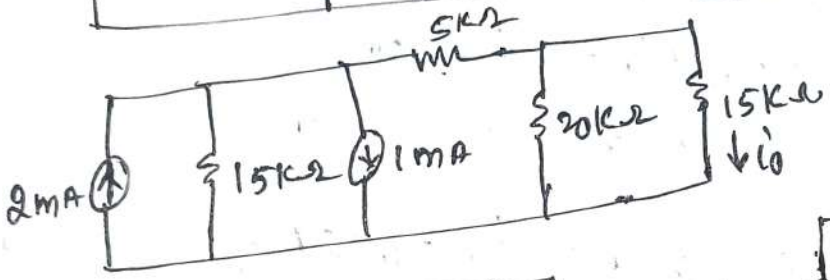
Q.3 :

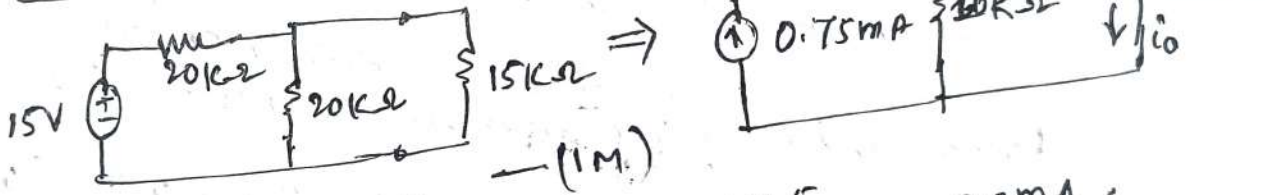




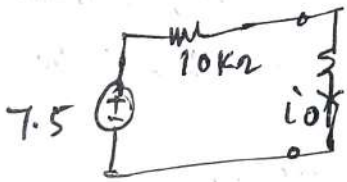


(1M)





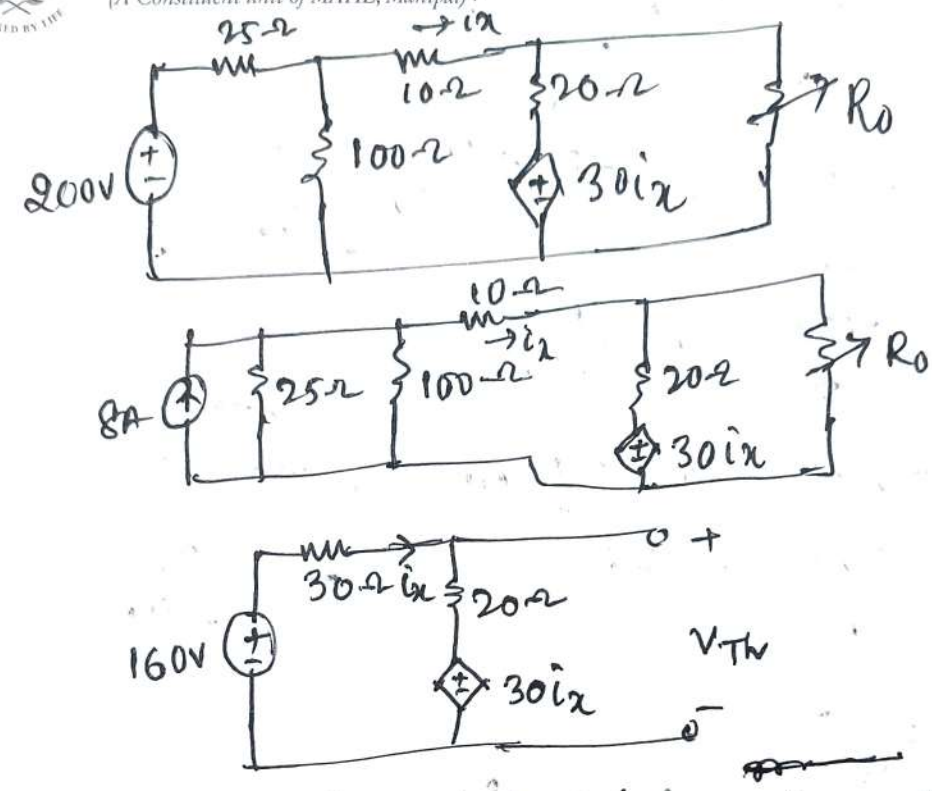
(1M)



$\therefore i_o = \frac{7.5}{10K+15K} = 0.3mA$

(1M)

Q.4

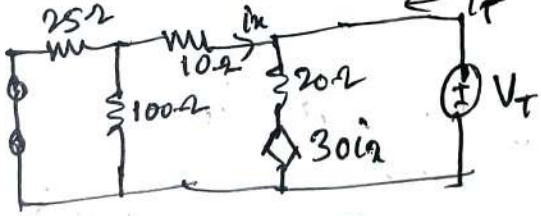


$$i_x = \frac{160 - 30i_x}{50} \Rightarrow i_x = 2A \quad \text{—— (1M)}$$

$$\therefore V_{Th} = 20i_x + 30i_x$$

$$= 50i_x = 100\ V.$$

(please consider other method also)

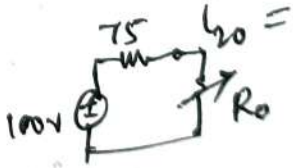


$$i = \frac{V_T}{25+10} = \frac{V_T}{35}$$

$$i_T = \frac{V_T}{30} - \frac{V_T - 30(-V_T/30)}{20}$$

$$\frac{i_T}{V_T} = \frac{1}{30} + \frac{1}{10} = \frac{2}{15}$$

$$R_{Th} = \frac{V_T}{i_T} = 7.5\Omega \quad \text{—— (2M)}$$



$$i_{20} = \frac{V_T - 30i_x}{20}$$

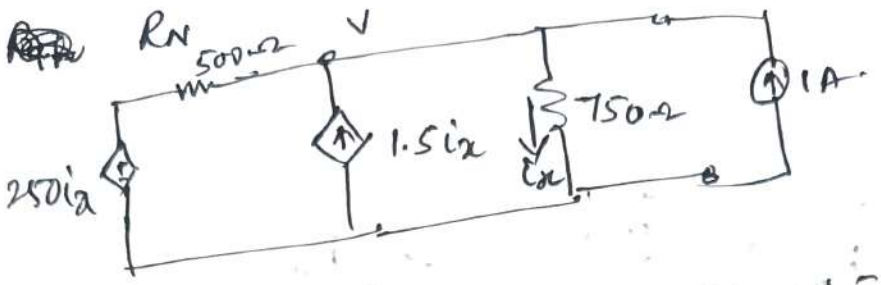
$$\therefore P = \left(\frac{100}{7.5 + R_o}\right) R_o = 250$$

$$R_o = 12.5 \pm 10$$

$$= 22.5\Omega$$

$$= 2.5\Omega \quad \text{—— (1M)}$$

Q5: $V_{th} = 0$, since no independent source

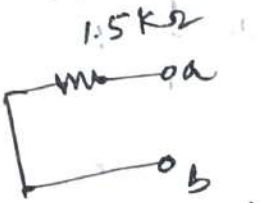


$$\frac{V - 250 i_x}{500} - 1.5 i_x + \frac{V}{750} - 1 = 0 \quad \text{— (1M)}$$

but $i_x = \frac{V}{750}$

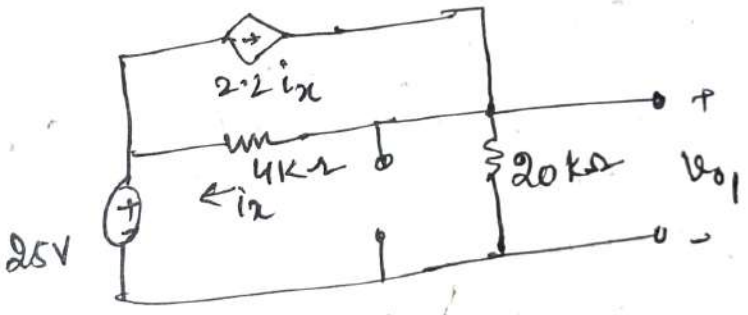
$\therefore v = 1500V$; $i_x = 2A$ — (1M)

$\therefore R_N = \frac{1500}{1A} = 1.5K\Omega$.



— (1M)

Q6:



KCL

$$\frac{V_{o1} - 25}{4K} + \frac{V_o}{20K} - 2.2 i_x = 0$$

$$\frac{V_{o1} - 25}{4K} + \frac{V_o}{20K} - 2.2 \left(\frac{V_{o1} - 25}{4K} \right)$$

$$5V_{o1} - 125 + V_{o1} - 11V_{o1} + 275 = 0$$

$V_{o1} = 30V$ — (1M)

Current source acting alone

$$\frac{V_{o2}}{4000} + \frac{V_{o2}}{20K} + 0.005 - 2.2 \left(\frac{V_{o2}}{4K} \right) = 0$$

$$5V_{o2} + V_{o2} + 100 - 11V_{o2} = 0$$

$\therefore V_{o2} = 20V$ — (1M) $V_o = V_{o1} + V_{o2} = 50V$ — (1M)

Q87: $v(0^-) = \frac{5}{5+3} 24 = 15\,V$ — (1M)

$v(0^-) = v(0^+) = 15\,V$

$R_{Th} = 4\,K\Omega$ — (1M)

$\tau = R_{Th} \cdot C = 2s$

$\therefore v(t) = v(\infty) + (v(0) - v(\infty)) \cdot e^{-t/\tau}$

$\therefore v(0) = 15\,;\ v(\infty) = 30$ — (1M)

$v(t) = 30 + (15 - 30) e^{-t/\tau} \Rightarrow (30 - 15 e^{-0.5t})\,V$

@ $t = 1$, $v(1) = 30 - 15 e^{-0.5 \cdot 1} = 20.902\,V$

@ $t = 4$, $v(4) = 30 - 15 e^{-(0.5 \times 4)} = 27.97\,V$ — (1M)

Q88:

$v_c(t) = \frac{9}{9+3}(20) = 15\,V\,;\ t < 0$

$v_c(0) = V_0 = 15\,V$ — (1M)

$R_{eq} = 1 + 9 = 10\,\Omega\,;\ t > 0$

$\tau = R_{eq} C = 10 \times 20 \times 10^{-3} = 0.2s$ — (1M)

$v(t) = V_c(0) e^{-t/\tau} = 15 e^{-t/0.2}\,V$

$\therefore$ energy stored in capacitor is

$w_c(0) = \frac{1}{2} C v_c^2(0) = 2.25\,J$ — (1M)

Q89: $v_c(0^-) = 40V$ @ $t = 0^-$

$i(0^-) = 0$

@ $t = 0^+$

$v_c(0^+) = 40V$

$i(0^+) = 0$ —(1M)

@ $t > 0$

$$-1\frac{di}{dt} - 20i - \frac{1}{1\times10^{-6}}\int_0^t i\,dt - 40 = 0$$

$$\frac{di}{dt}(0^+) = -40\ A/s$$

$$\frac{d^2i(0^+)}{dt^2} = 800\ A/s^2$$

—(2M)